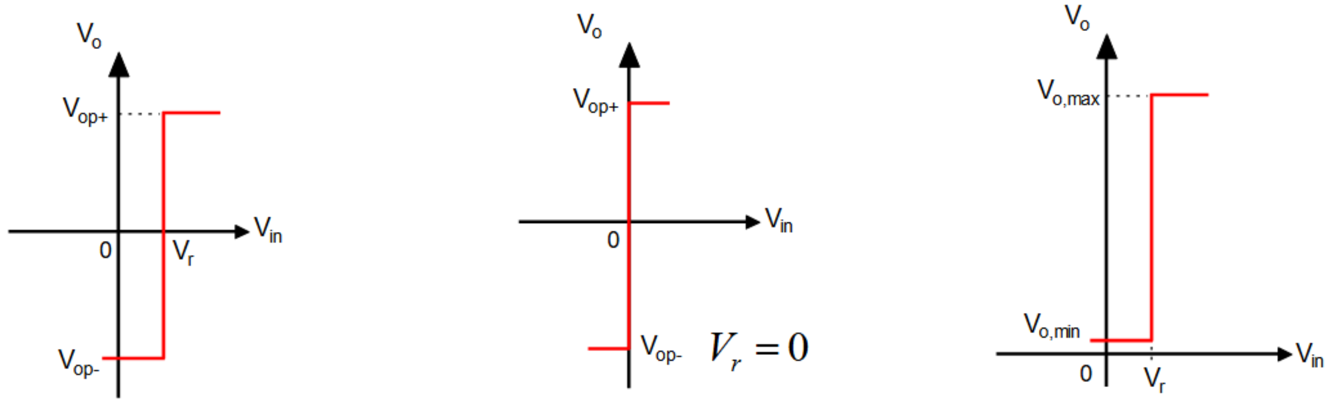
Comparators

Comparator Transfer Characteristics

Ideal



$$V_0 = \begin{cases} V_{op+} \leftarrow (V_{in} > V_r) \\ V_{op-} \leftarrow (V_{in} < V_r) \end{cases}$$

(Compare input with V_r)

$$V_0 = \begin{cases} V_{op+} \leftarrow (V_{in} > 0) \\ V_{op-} \leftarrow (V_{in} < 0) \end{cases}$$

($V_r = 0$)

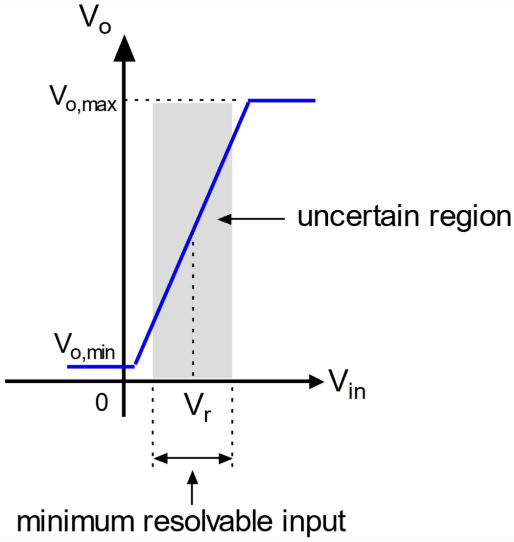
$$V_0 = \begin{cases} V_{o,\max} \leftarrow (V_{in} > V_r) \\ V_{o,\min} \leftarrow (V_{in} < V_r) \end{cases}$$

(Single supply)

※ For realisation, it is desired to have high gain, high speed, low offset and hysteresis

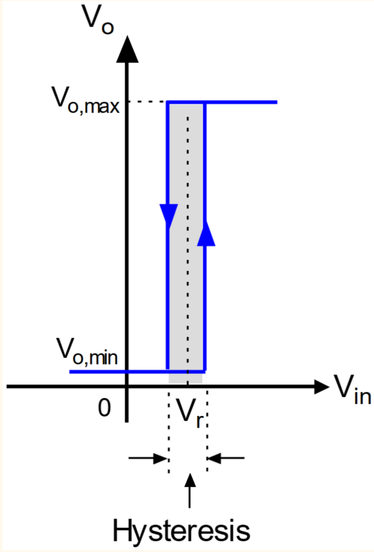
Non-Ideal

Finite gain



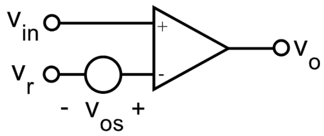
※ The uncertain region will give the minimum resolvable input, the minimum difference V_{in} must be from reference

Hysteresis

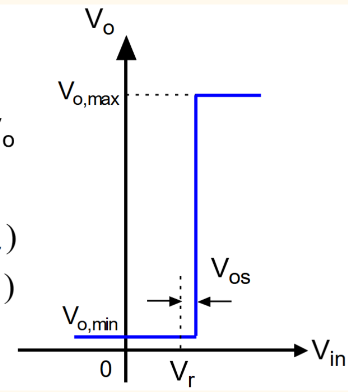


※ Hysteresis is not always bad ⇒ can prevent noise causing multiple triggers when its near the threshold
※ But its unwanted for a pure comparator

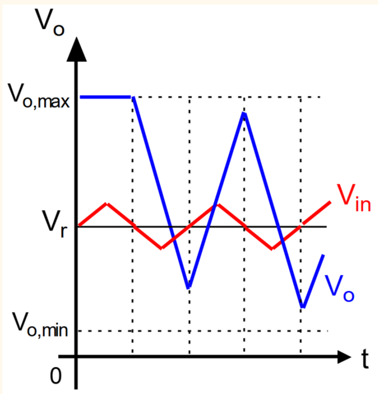
Offset



$$V_0 = \begin{cases} V_{o,\max} \leftarrow (V_{in} > V_r + V_{os}) \\ V_{o,\min} \leftarrow (V_{in} < V_r + V_{os}) \end{cases}$$

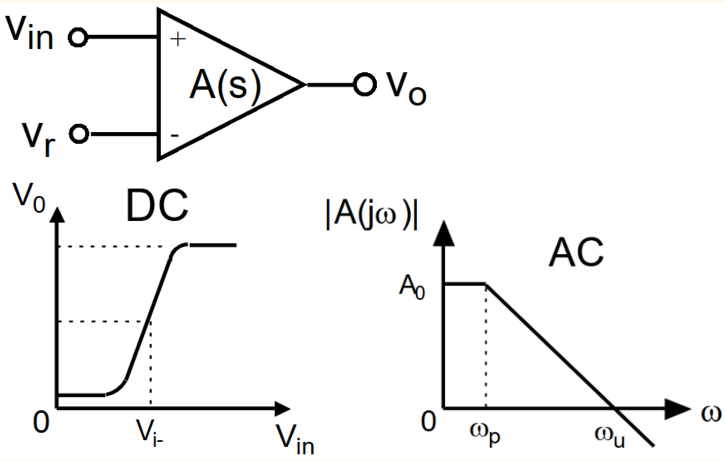


Slew rate limit



Circuit Realisation

High Gain Op Amp



※ For op amp, to get high gain means its open loop, but the open loop BW is typically very small, so the settling time is very slow.
⇒ Only good for low speed application
※ To decrease the settling time, the BW can be extended with feedback, but the gain is reduced.
※ If its not slew rate limited:

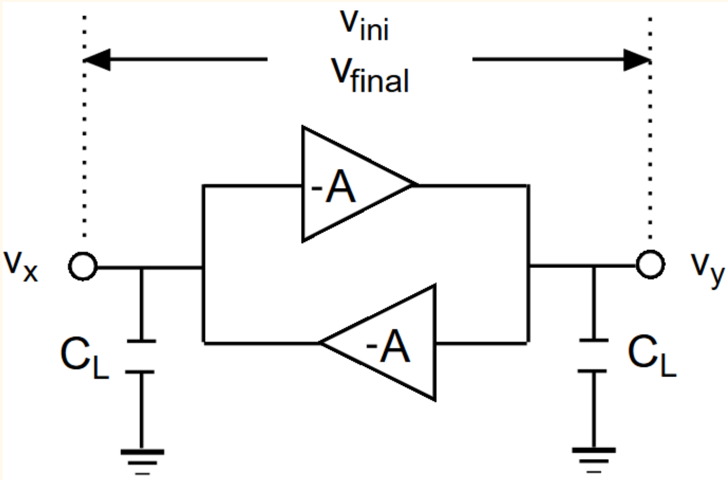
$t_{settling,\,1\%} \approx \frac{4.6}{\omega_{CL,\,3dB}}$ $\quad t_{settling,\,0.1\%} \approx \frac{6.9}{\omega_{CL,\,3dB}}$

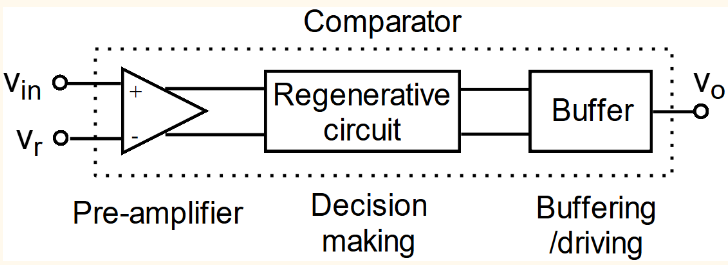
$t_{settling,\,x\%} = \frac{1}{\omega_{CL,\,3dB}} ln(\frac{1}{x/100})$

※ If its slew rate limited:

$t_{settling} \approx \frac{\Delta V}{SR} \frac{6.9}{\omega_{CL,\,3dB}}$ $\quad SR = \frac{I}{C_L}$

Regenerative (or Latch) Comparator





※ For comparator, it is not desired to have linear settling time or stability, we want the transition to be fast ⇒ positive feedback is used

※ Two inverting amplifier looping each other is called a regenerative latch

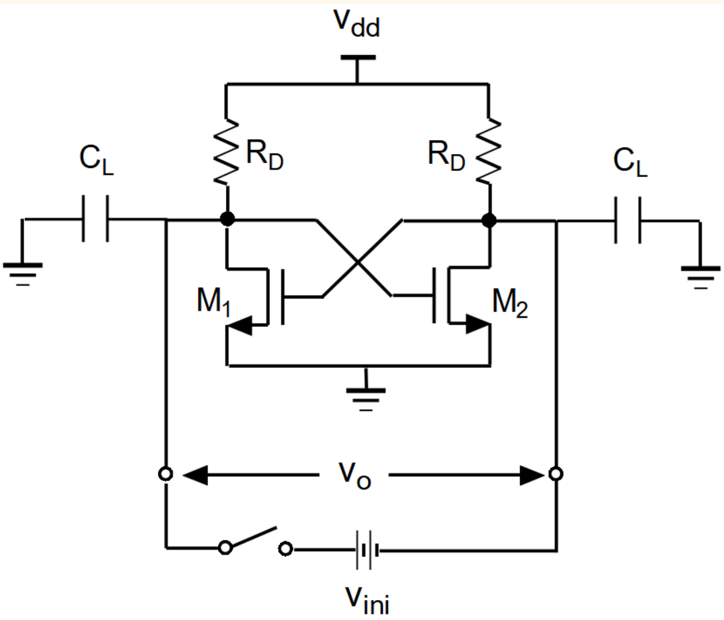
- if one side is more positive, it makes the other side more negative, which makes the first side more positive

※ The input and output ports are the same

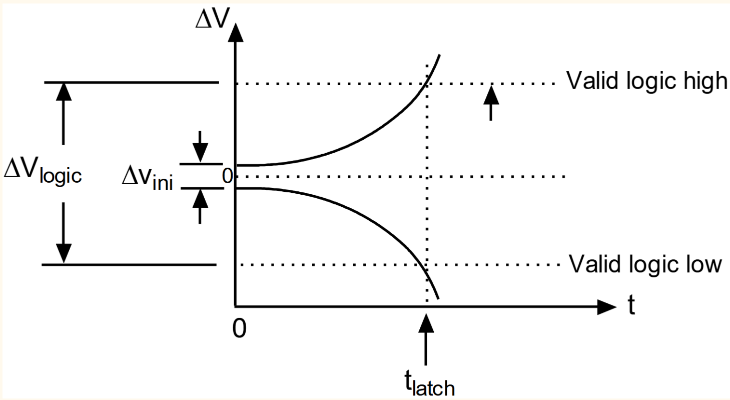
※ The general topology of comparator is a preamp, the latch circuit and some buffer

※ The initial voltage at the certain polarity is needed, which is given by the preamp.

- think of ΔV_{ini} as the direction we want the latch to move towards after we enable it
- see [Designs Literature Review]



※ This kind of criss cross inverter circuit is called a latch circuit

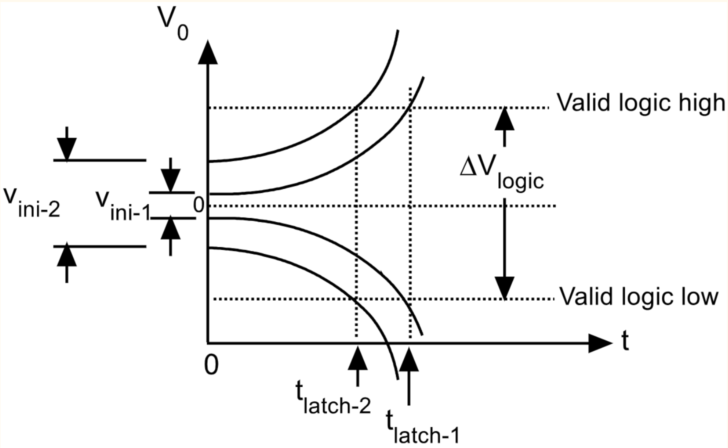


$\Delta V = V_x - V_y$ $\quad \Delta V \approx \Delta V_{ini} e^{\frac{g_m}{C_L}t}$ $\quad t_{latch} \approx \frac{C_L}{g_m} ln(\frac{\Delta V_{logic}}{\Delta V_{ini}})$

※ Expression are derived from transient response and solving the differential equation. The voltage will increase exponentially in time.

※ Latching time is the time taken for the circuit to reach ΔV_{logic} with a initial voltage of ΔV_{ini}

Metastability Problem in Latch



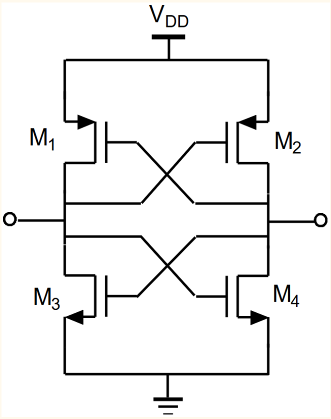
※ Output of comparator are read every clock cycle. If the latching time is too long, the output may not reach logic levels before it is read, making it invalid. This uncertainty in reading is called metastability

※ To avoid metastability ⇒ $t_{latch} \leq T_{clock}$

$$t_{latch} \approx \frac{C_L}{g_m} ln(\frac{\Delta V_{logic}}{\Delta V_{ini}})$$

※ To reduce metastability ⇒ ↓ t_{latch}:

1. Increase ΔV_{ini} by adding preamp or cascade latches (high gain, bigger difference)
2. Decrease C_L or increase g_m



$C_L \approx 4WLC_{ox}$ (Assume driving another inverter, M5, M6)

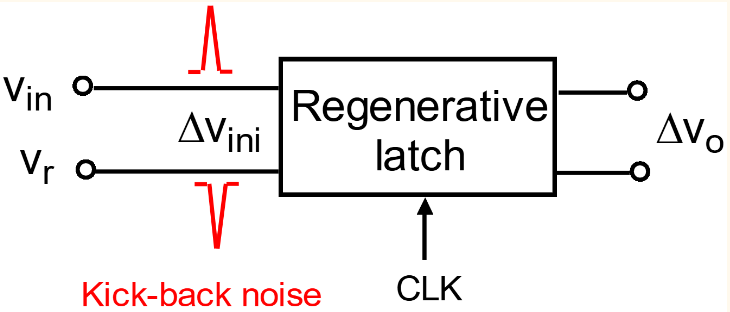
$$g_m \approx \mu C_{ox} \frac{W}{L}(V_{gs} - V_T)$$

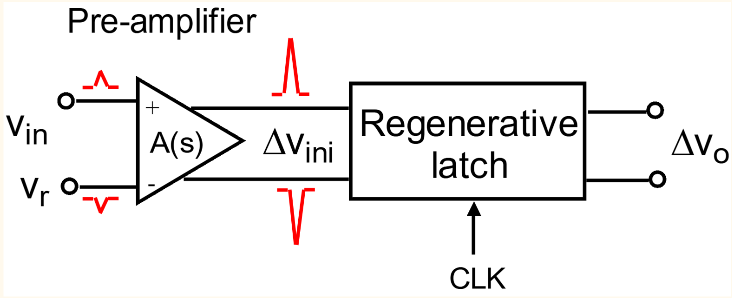
$$\frac{C_L}{g_m} \approx \frac{4WLC_{ox}}{\mu C_{ox} \frac{W}{L}(V_{gs} - V_T)} = \frac{4L^2}{\mu(V_{gs} - V_T)}$$

※ C_L/g_m is mostly technology/process dependant

※ only can change V_{GS} by changing the current

"Kick-back" noise





※ Fast change in output voltage can coupled back to the input by through coupling (e.g. C_{gd} C_{gs})

※ How big of a noise induced depends on the impedance of the previous stage

- using preamp helps to reduce the impedance and reduce the noise
  - reduced impedance means the noise transfer function is smaller gain
  - having a preamp increases the number of coupling the noise need to jump across, hence its attenuated
- preamp can both reduce kick back and give the initial voltage difference

Designs Literature Review

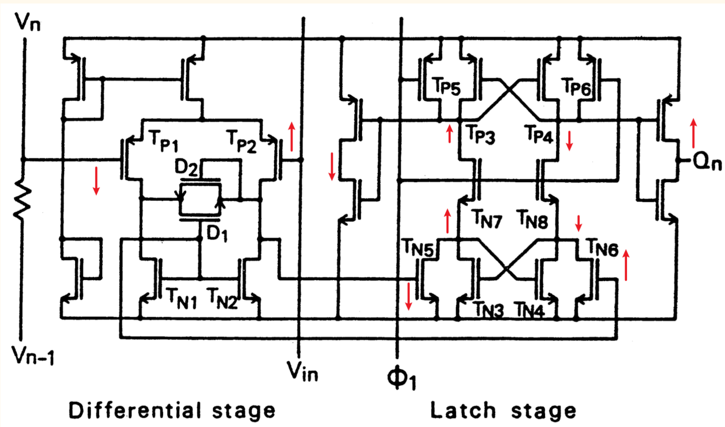
Common observation between design 1 and 2, during the track phase

- The output is reset and the interaction between the PMOS and NMOS latches is broken by some switch in between.
- There is some ΔV_{ini} to be applied or already applied by the differential stage
- Once the prohibition/reset signal is removed the comparator will swing towards the direction ΔV_{ini} is point towards

Design 1

Yukawa, A. “A CMOS 8-bit High-Speed A/D Converter”, IEEE JSSC, June 1985

- preamp → latch



Arrows in red shows one of the possible latch direction (Size of arrow has no meaning)

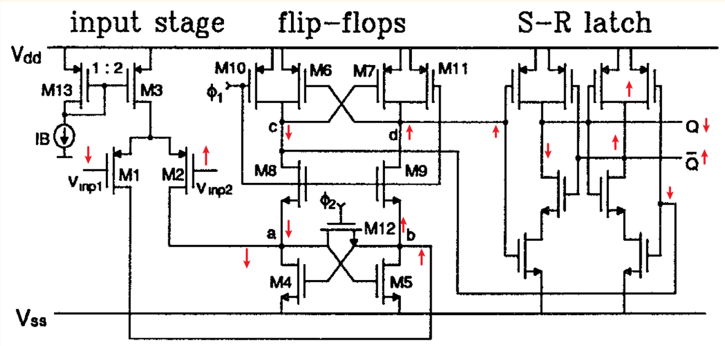
※ PMOS NMOS diode in series is to create some voltage reference, the VTHN and VTHP variation cancels out. See here.
※ D1 D2 is to limit the differential pair output swing, so can recover faster
※ BW of preamp is limited by high R_{out}
※ $A_{V, preamp} = g_{m, P2}\,(r_{o, P2} \,||\, r_{o, N2})$
※ No static current in latch ⇒ low power consumption
※ The differential pair is a single ended configuration with the NMOS active load, this means the T_{N5} T_{N6} are driven asymmetrically. But it's ok, since it's a latch.
※ Kick back noise reduced by preamp because from output need to jump two C_{GD} to reach input. The kick back noise will be asymmetrical, since one side of the NMOS active load is diode connected. But it should be small enough to ignore anyways.

Track Phase: $\phi_1 = 0$	Latch Phase: $\phi_1 = 1$
※ Drain voltage of T_{P3} and T_{P4} are at V_{DD} ※ Drain voltage of T_{N3} and T_{N4} are at 0 ※ There is no current flow ※ $Q_n = 0$	※ T_{N7} and T_{N8} can flow current and the drain voltage of T_{N3} and T_{N4} can change ※ The ΔV_{ini} developed depends on the output of the differential stage, then the regenerative process (from TP3, TP4, TN3 and TN4) will bring it to completion

Design 2

Yin, G M, et al "A High-Speed CMOS Comparator with 8-bResolution", IEEE JSSC, February 1992

- preamp → cascade latches



Arrows in red shows one of the possible latch direction (Size of arrow has no meaning)

※ M4 M5 are acting as preamp and latch

※ $A_{V,\,preamp} = \frac{1}{2} g_{m1,2} R_{on,\,12}$

※ Kick back noise could be high because the preamp output impedance is high (other designs are diode connected) and only need to jump 1 C_{GD}

※ SR latch is 2 inverters back to back. It is added so that the comparator 1st decision is held until the 2nd decision is made. Output does not change when the comparator is changing.

c	d	Q+
1	1	Q
0	1	0
1	0	1
0	0	NA

EE2026 Notes

Track Phase: $\phi_1 = 0$, $\phi_2 = 1$

※ Votlage at c and d are pulled to V_{DD} by M10 and M11

※ M8 M9 doesn't conduct

※ Differential current from preamp is developing ΔV_{ini} across $R_{on,\,12}$

※ SR Latch is holding the previous decision

Latch Phase: $\phi_1 = 1$, $\phi_2 = 0$

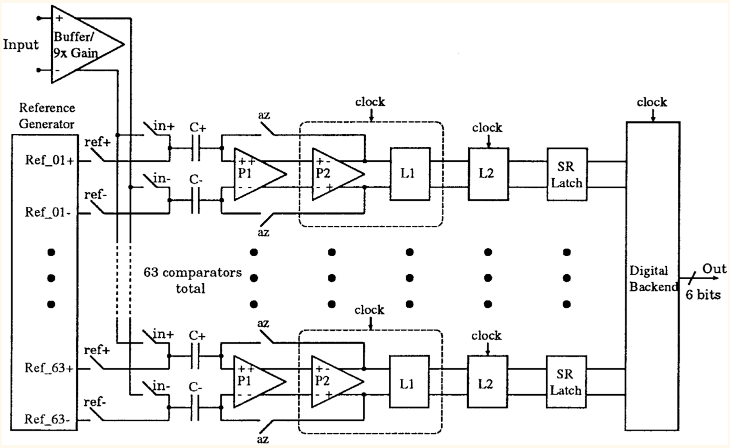
※ M10, M11 and M12 turn off, M8 M9 connect the PMOS and NMOS latch together

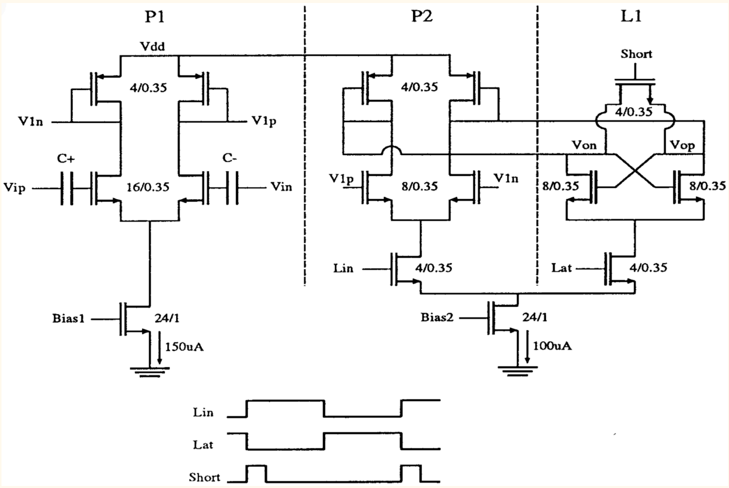
※ The regenerative process (from M4, M5, M6 and M7) will bring it to completion

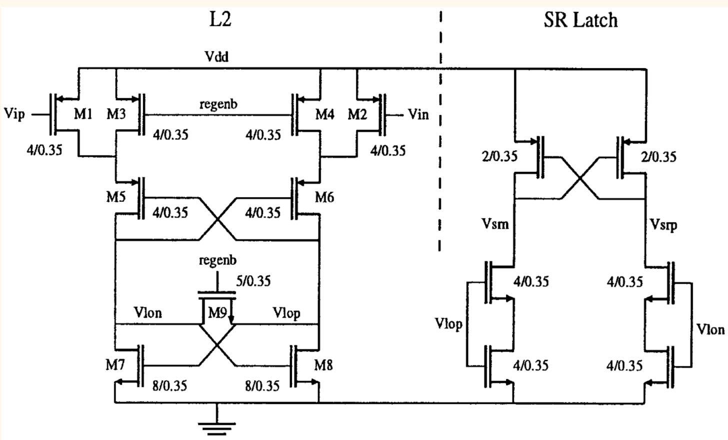
Design 3

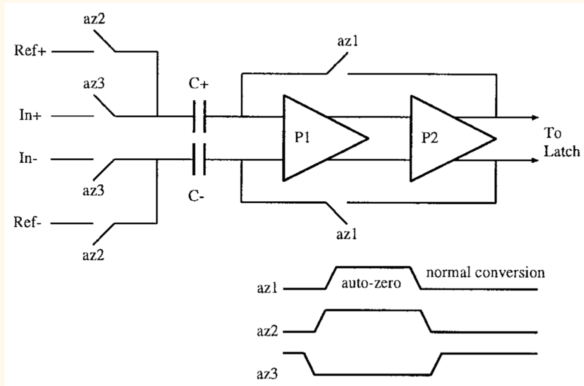
Mehr I, “A 500-Msample/s, 6-bit Nyquist-rate ADC for Disk-Drive Read-Channel Applications”, IEEE JSSC, July 1999

- 2 preamp, 2 cascade latches, plus a SR latch









※ The first stage does auto offset zeroing

※ More here

==

※ Flash ADC with 63 comparator for 6-bit output

※ Each preamp is a diode connected load differential pair. Cascading amplifiers like this can increase gain while maintaining BW (no need to trade off with increasing R_{out}).

⇒ increase initial voltage and reduce kick back noise

※ Having 3 latches reduces metastability

※ Same current source is shared between P2 and L1

Phase 1: Lin = 1, Lat = 0, Short = 1

※ P2 is enabled, L1 is disabled

※ Short resets the latch and shunts the P2 output

Phase 2: Lin = 1, Lat = 0, Short = 0

※ Allow ΔV_{ini} to develop across L1

Phase 3: Lin = 0, Lat = 1, Short = 0

※ P2 is disabled, L1 is enabled

※ NMOS latch is form with PMOS diode connected load from P2

※ The regenerative process will bring it to completion

Comparator Offset Cancellation (Auto Zeroing)

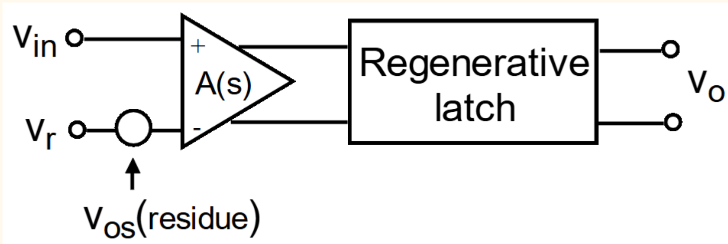
※ With proper designs, the mismatch should be dominated by the input devices, i.e., the input differential pair are the preamp

※ **Strategy: store the offset at some capacitor at the initial clock phase (sample the offset first), then during the usage phase, the offset is subtracted out**

※ The caveat with using such sampling switch is charge injection:

- When conducting, MOS has a channel.
- When the channel is turn offed, the charge/carrier in the channel has to go somewhere, either to D or to S
- If the charge exiting is entering into a cap, then the charge is injecting into the cap, changing the voltage
- This causes error in the offset voltage to be subtracted later on, causing some left over offsets

※ There is also charge absorption, which is when channel is turned on, taking charges from cap



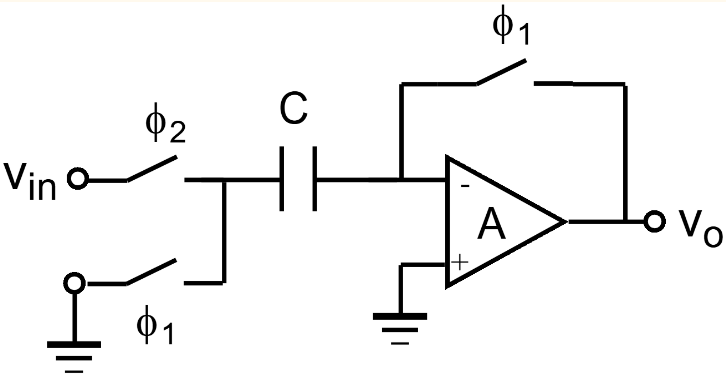
※ The latch itself, also has some offset, V_{osl}. But it's reduced by A when referred to input

※ In general, after offset cancellation, the input residual offset is dependant on:

- Leftover offset in A after cancellation
- Charge injection into cap, $\Delta Q/C$
- Latch offset, V_{osl}

* Remember to divide by A, if its on the output

Cancellation at Input (Closed-Loop Cancellation):



※ Here, the offset is stored in C at the input

※ During sampling phase:

$$V_C = \frac{A}{1+A} V_{OS} \quad \leftarrow \text{not } V_{OS} \text{ since } A \neq \infty$$

※ During sampling phase:

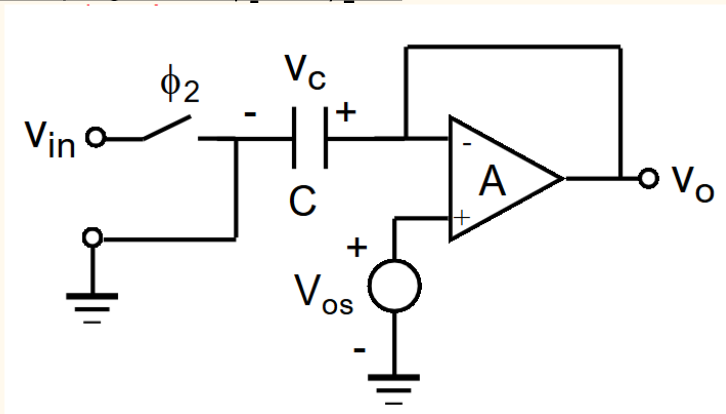
$$V_O = -A(V_{in} - V_C) + AV_{OS} = \frac{A}{1+A} V_{OS} - AV_{in}$$

※ Input referred residua offset:

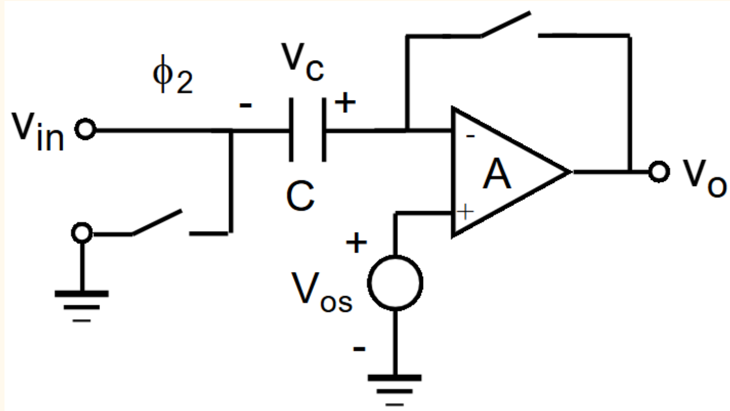
$$V_{OS,\,residue} = \frac{1}{1+A} V_{OS} + \frac{\Delta Q}{C} + \frac{V_{osl}}{A}$$

※ To reduce residue offset, need to increase A

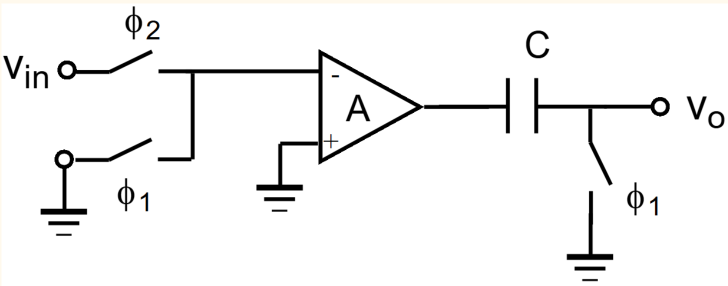
Sampling Phase: $\phi_1 = 1$, $\phi_2 = 0$



Usage Phase: $\phi_1 = 0$, $\phi_2 = 1$



Cancellation at Output (Open-Loop Cancellation):



※ Here, the offset is stored in C at the output

※ During sampling phase:

$$V_C = AV_{OS}$$

※ During sampling phase:

$$V_O = -AV_{in} + AV_{OS} - V_C = -AV_{in}$$

※ Input referred residua offset:

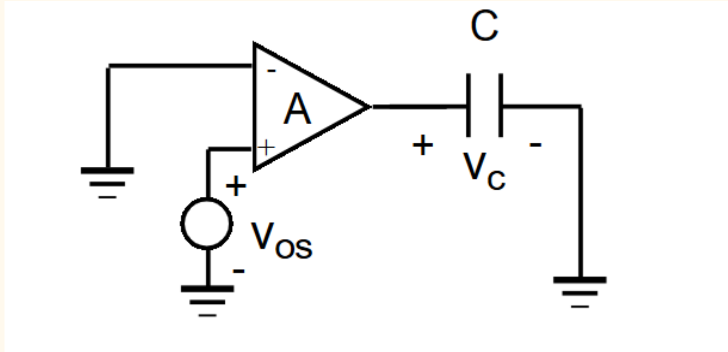
$$V_{OS,\,residue} = \frac{1}{A}\frac{\Delta Q}{C} + \frac{V_{osl}}{A}$$

※ Complete offset cancellation is achieved without needing A to be big

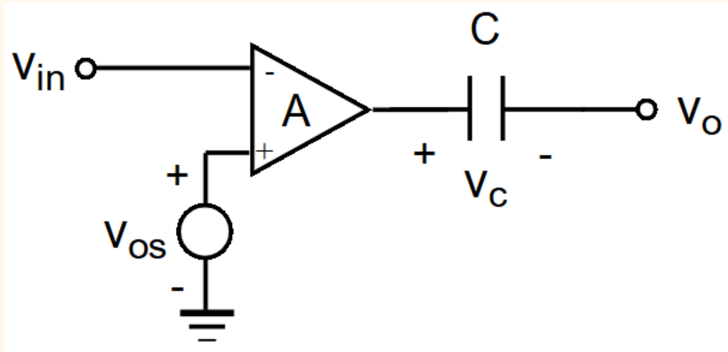
※ Offset stored in C is offset x A, where A is the open loop gain of preamp. To ensure the output doesn't get clipped, the gain is limited.

※ Preamp will need to drive a capacitance load

Sampling Phase: $\phi_1 = 1$, $\phi_2 = 0$



Usage Phase: $\phi_1 = 0$, $\phi_2 = 1$



Designs Literature Review

- Some call the auto zero phase calibration

Design 1

Dingwall, “Monolithic expandable 6 bit 20MHz CMOS/SOS A/D converter,” JSSC 1979

Dingwall, “An 8-MHz CMOS subranging 8-bit A/D Converter,” JSSC 1985

- SOS means silicon on sapphire

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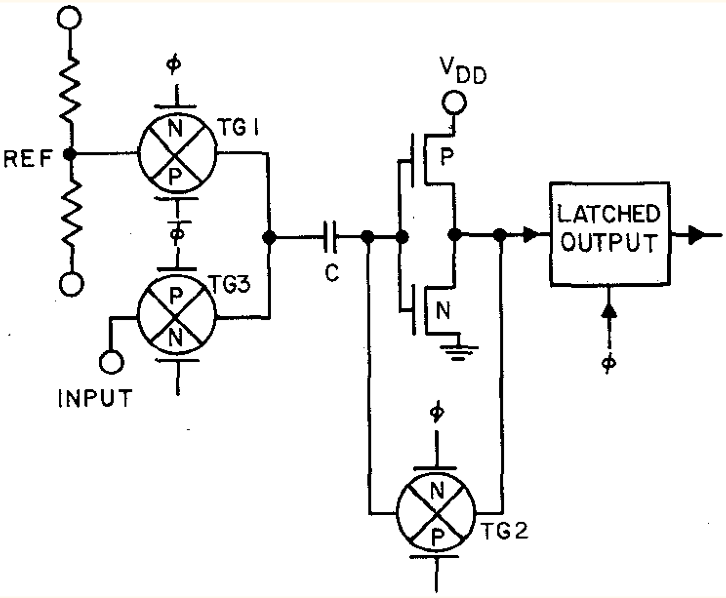

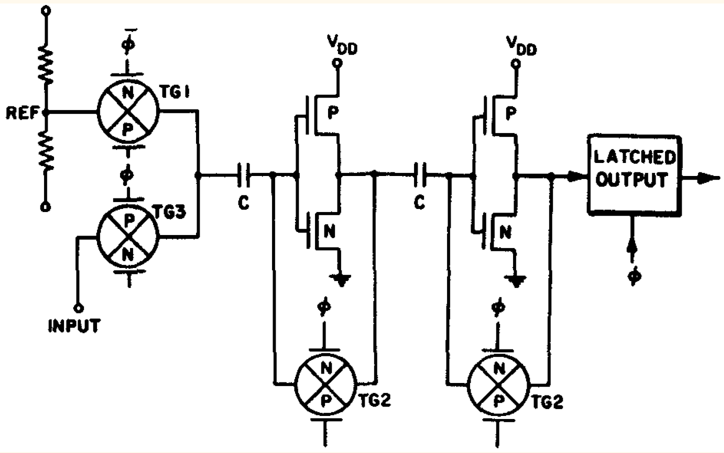

C used is 10x larger than parasitic
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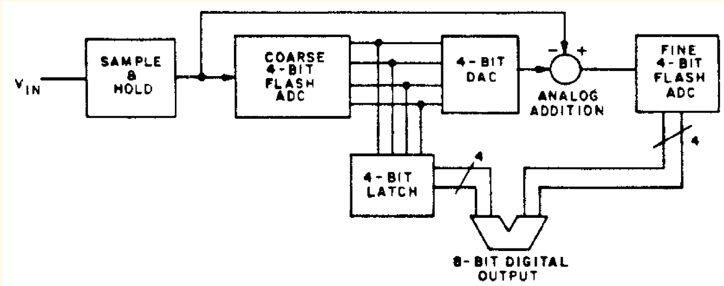
※ Inverter as inverting amp is popular because its a simple circuit
※ Both of these are offset cancellation at the input


※ Subranging means cut the bit depth and do the big and small values desperately (the number of comparators needed in the flash ADC is reduced alot)
※ See this doc from ADI
==
<u>ϕ1: TG1 = 1, TG2 = 1, TG3 = 0:</u>
※ TG2 will make the inverter autozeroes to its toggle point (the voltage at the input that makes the output have the same voltage)
※ TG1 cause C to hold V_{ref} - V_{toggle}

<u>ϕ2: TG1 = 0, TG2 = 0, TG3 = 1:</u>
※ TG3 shift the left of C by V_{in}, any difference between V_{in} and V_{ref}, will amplified by the inverter gain
※ The latch circuit will make the output a valid digital high or low
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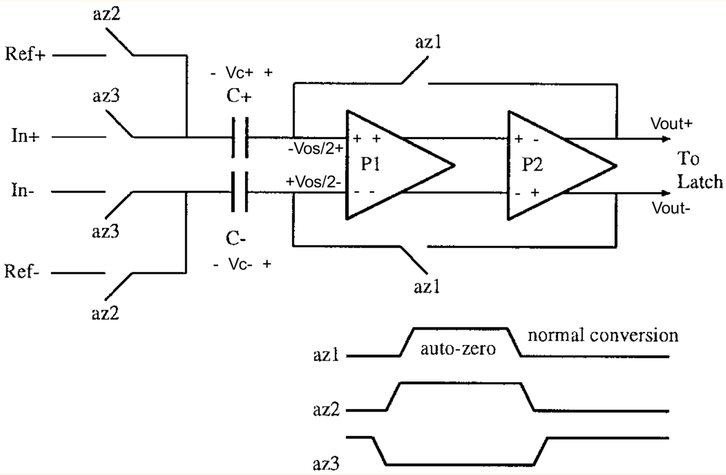
Design 2

Mehr I, “A 500-Msample/s, 6-bit Nyquist-rate ADC for Disk-Drive Read-Channel Applications”, JSSC, July 1999



※ az2 and az3 are implemented with NMOS since the voltage transmitted is closer to GND than VDD

※ az1 is implemented with PMOS since the voltage is closer to VDD than GND

※ With cascaded opamps, the gain is high, so cannot do output cancellation. This is also input cancellation

==

<u>ϕ1: az1 = 1, az2 = 1, az3 = 0:</u>

$$V_{out} = -\frac{A_1A_2}{1+A_1A_2}V_{OS}$$

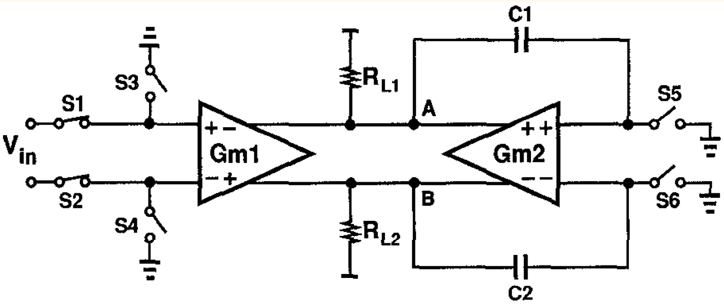
$$V_C = V_{C+} - V_{C-} = -\frac{A_1A_2}{1+A_1A_2}V_{OS} - V_{ref}$$

<u>ϕ2: az1 = 0, az2 = 0, az3 = 1:</u>

$$V_O = -A_1A_2(V_{in} - V_{ref}) - \frac{A_1A_2}{1+A_1A_2}V_{OS}$$

Design 3

Razavi, “Design techniques for high-speed high resolution comparators”, JSSC 1992

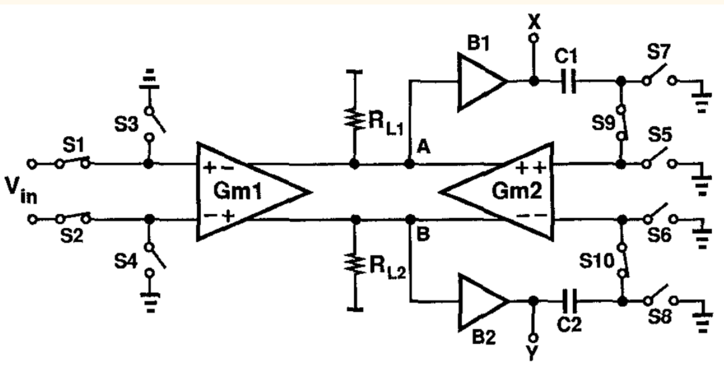


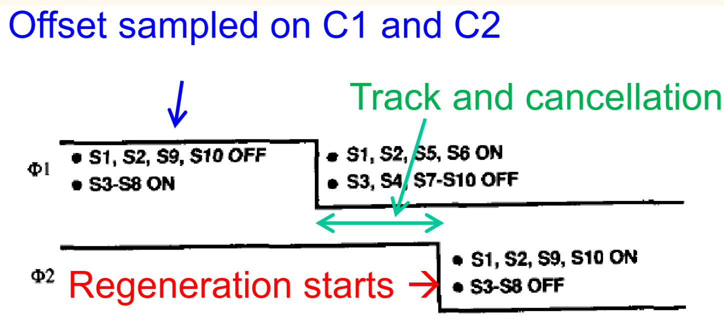
※ Two transconductance amplifier sharing output node
※ R_L is load resistance
※ C gives positive feedback loop around G_{m2}

<u>ϕ1: S1, S2 = 0, S3, S4, S5, S6 = 1</u>
※ Offset of both G_{m1} and G_{m2} are amplified and stored in C1/C2

<u>ϕ2: S1, S2 = 1, S3, S4, S5, S6 = 0</u>
※ Input is amplified by G_{m1} to establish an imbalance at A and B (ΔV_{ini}), which initiates regeneration around G_{m2}
※ Complete cancellation of both offsets is achieved. Unlike conventional output offset storage circuit that requires another latch circuit with offset

But this circuit would not work in practice:
※ C1 C2 and their parasitic load the output node, reducing speed
※ Finite on resistance of S5 and S6 means during calibration, the positive feedback loop is not completely broken, ⇒ prone to oscillation
※ Any mismatch in charge injection by S5 and S6 may cause regeneration to go other other way if the ΔV_{ini} is small ⇒ large input referred offset





※ B1 B2 added to isolate output nodes A and B from the feedback capacitor C1 C2. Any offset from B1 and B2 are also stored in C1 and C2 and cancelled later.
※ S7-S10 can control when exactly feedback loop is disabled. So, regeneration begins only when input voltage has been sensed and amplified
※ Residual offset is due to S7-S10 mismatch in charge injection and absorption
==

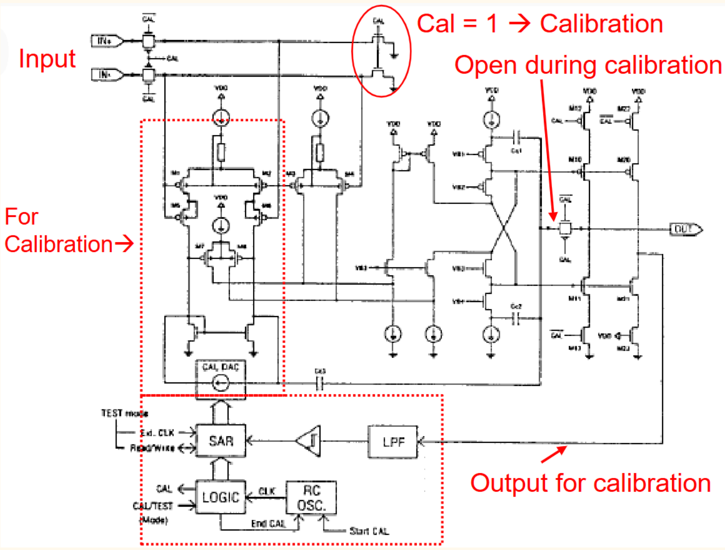
<u>ϕ1:</u>
※ Offset of G_{m1}, G_{m2}, B1 and B2 are amplified and stored in C1/C2

<u>ϕ2:</u>
※ Input is amplified by G_{m1} to establish an imbalance at A and B (ΔV_{ini}). But no regeneration occurs, since positive feedback loop is still broken by S9 S10

<u>ϕ3:</u>
※ Regeneration around G_{m2} is started

<u>Design 4</u>

Krummenacher, "A high-performance autozeroedCMOS opamp with 50uV offset" ISSCC, 1997



The Cc are for compensation, and are taken out during calibration

※ Uses digital circuit to calibrate out the offset

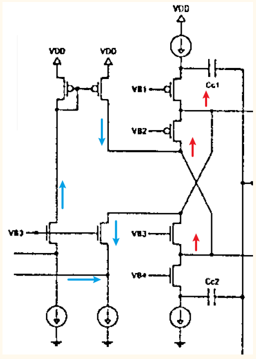
※ The left differential pair is a cascoded, higher amplifier r_o increases CMRR

※ A multipath nested Miller topology is used for frequency compensation.The additional input pair M3 and M4 (right) bypasses the intermediate stage M7 and M8 (left) at high frequencies to improve stability.

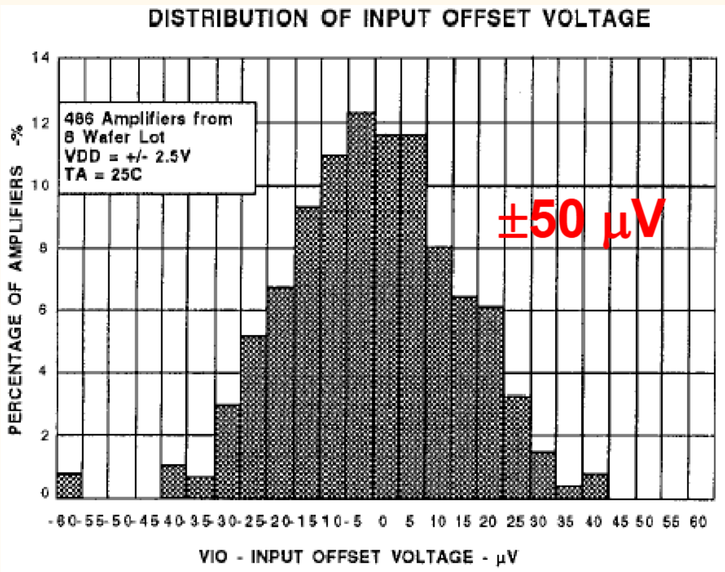
※ To transition from high-gain, low-frequency path to the low-gain, high-frequency path smoothly

⇒ $g_{m1,2}$ must match $g_{m3,4}$

※ Two output stages, one for calibration one for actual output. The one for calibration is a smaller replica.



※ Blue is change in current, Red is change in voltage, the regenerative loop is within this section



※ Achieves µV levels of offset